

4(a)	any two from: presence of mucus can, hinder / prevent / AW, gas exchange / entry of air (to alveoli) / airflow (to alveoli) ; A named gases detail ; e.g. (lumens of) airways / respiratory bronchioles, are narrow or airways / respiratory bronchioles, are close to alveoli <i>suggestion that <u>respiratory</u> bronchioles have, few / no, cilia present so difficult to move mucus / AW ;</i> R bronchioles do not have cilia <i>as a general statement</i> mucus increases distance for diffusion of (respiratory) gases / oxygen / carbon dioxide ; R if direction of diffusion of gas is incorrect increased risk of infection from trapped pathogens if mucus present ; AW	2
4(b)(i)	any three from: (bronchioles) have columnar, epithelium / cells ; A columnar and cuboidal, epithelium / cells (bronchioles) have ciliated, epithelium / (epithelial) cells ; A cilia (are present) ora (alveoli) have squamous, epithelium / cells ; I endothelium / endothelial cells detail ; e.g. (bronchiole epithelial cells) longer / wider / thicker R if, mucus makes it thicker / stated as more than one layer or (alveolar cells) flatter / thinner <i>allow ideas if stated as epithelium (rather than cells)</i> AVP ; e.g. ref. to alveolar type II cells / cells secreting surfactant	3

Question	Answer	Marks
4(b)(ii)	<u>smooth muscle</u> ; plus any two from: max 1 for function if stretch and recoil or vasoconstriction / vasodilation stated (muscle) <u>contracts / contraction</u> (and relaxation of bronchiole, lining / wall) ; R contraction of, bronchiole / lungs / other airways change / decrease, diameter of, bronchiole / lumen / airway ; A adjust size of airway A constriction of, bronchiole / airway control of air flow (to and from alveoli) / AW ; e.g. control volume of air, entering / leaving, alveoli allow ecf max two marks if incorrectly stated as elastic fibres stretch / expand, during inhalation ; A stretch and recoil prevents, overstretching / bursting, during inhalation ; recoils to help move air out during exhalation; allow ecf to max one mark if incorrectly stated as cartilage prevent collapse of, bronchiole / airway ; provides support / keeps airway open ;	3